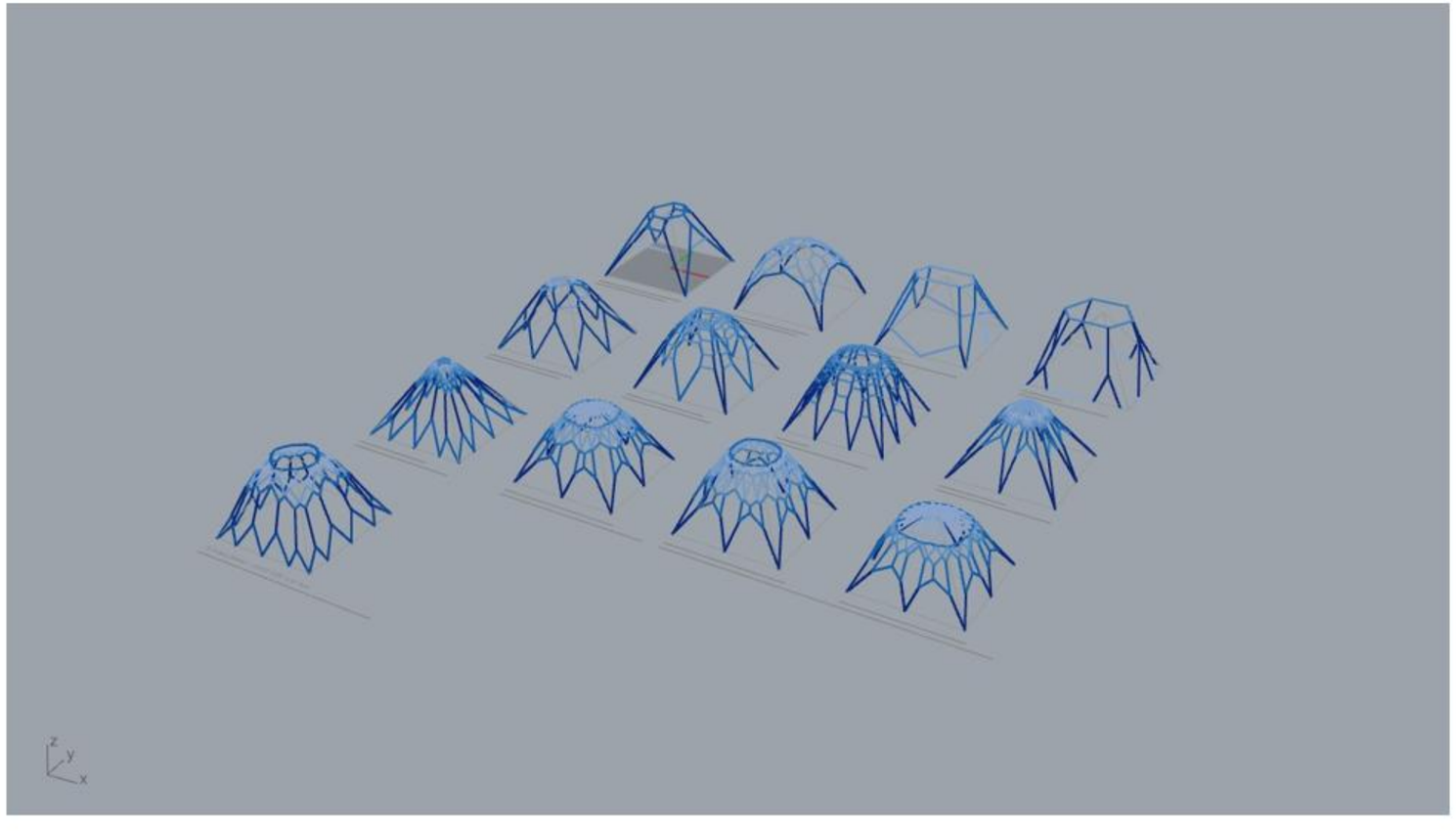
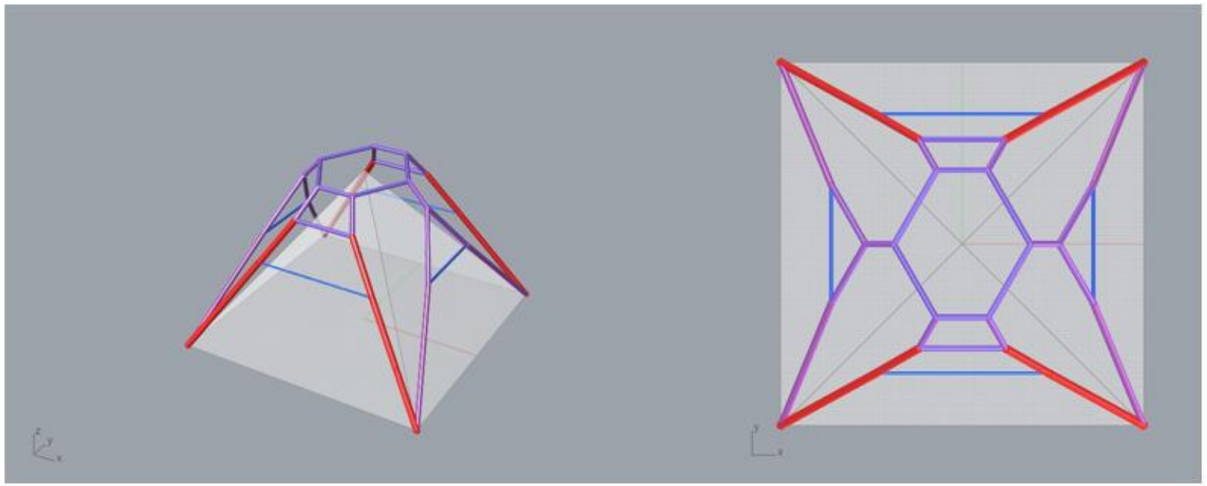
Friend's roof variants - all iterations (PolyFrame 2, 10x10 m plan, apex ~6.67 m)

All files in polyframe2 (not committed). 3D review: friend-iterations-review-v2.3dm, one layer per iteration. C = compression.
Karamba: cast C30/37, ribs 150x250, rigid joints, feet pinned, self-weight + 7 kN/m2 (assumed) on the pyramid surface; elastic stress check, no rebar.

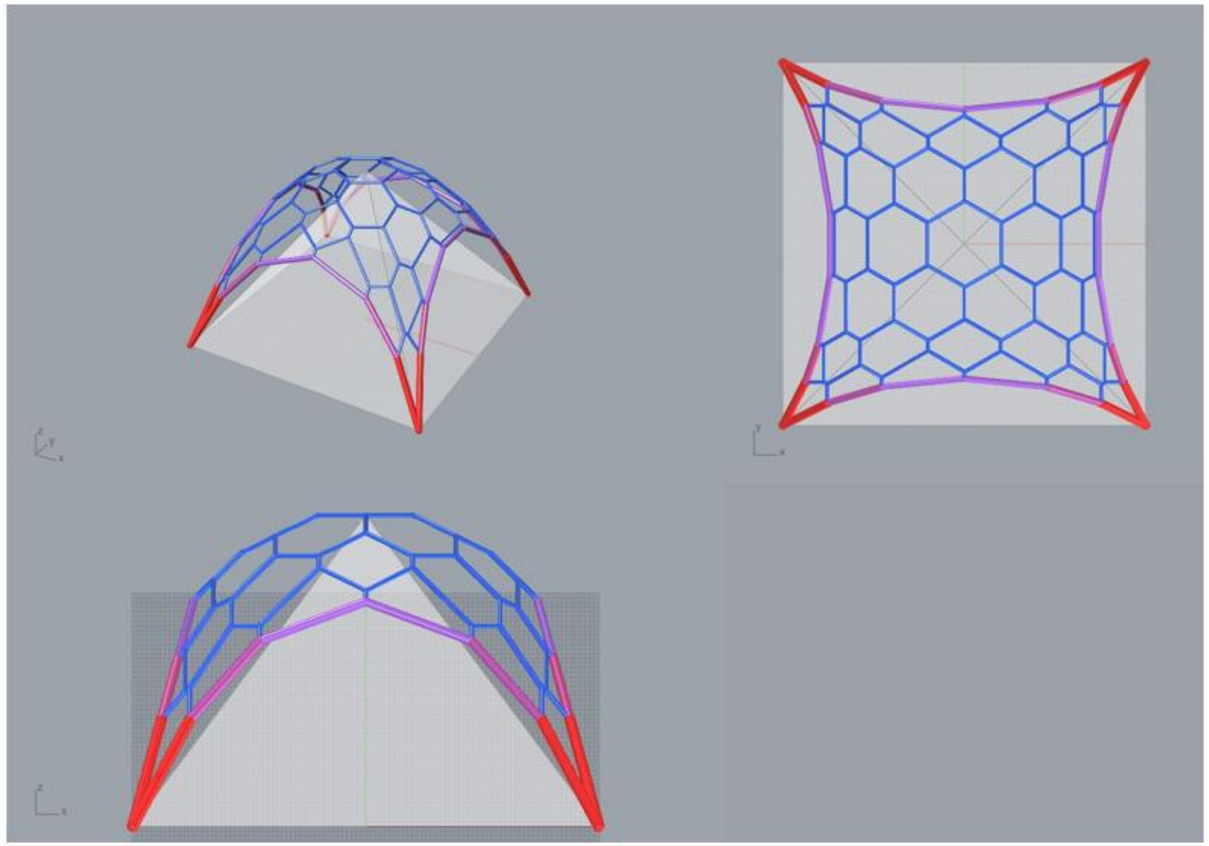
#	Iteration	Feet	PolyFrame final shape	Karamba comp/fcd	Bars <=35 deg	Known flaw	File
1	Panel A frame	4	exact, all C	-	-	not the sheet's look	friend-A-4corners
2	Panel A shell (ex 3(7))	4	exact, all C (136 bars)	-	-	free edges 1.3 m vs 2.3 m	friend-A-shell
3	AP8 frame	8	stage B all C	-	-	not the sheet's look	friend-AP8-buttresses
4	AP12 frame	12	stage B all C	-	-	not the sheet's look	friend-AP12-buttresses
5	AP8 shell GUESS	8	stage B all C	-	-	not symmetric	friend-AP8-shell-GUESS
6	AP8 shell v2 (rings)	8	stage B all C	-	24/248	96 horizontal ring bars	friend-AP8-shell-v2 (+printcast)
7	AP12 shell	12	stage B all C	-	112/516	rings horizontal; stage A tension	friend-AP12-shell
8	AP8 diagrid	8	stage B all C	1.03 (corner legs)	24/272	stage A 40 tension	friend-AP8-diagrid
9	**16-feet diagrid (5/side)**	**16**	**A and B all C**	**1.17 (mid-face)**	**16/192**	**mid-side feet idle**	**friend-16-diagrid**
10	AP8 even load (hub)	8	A and B all C	0.58	24/272	48 flat hub bars; ring overlap	friend-AP8-diagrid-evenload
11	**AP8 even load OPEN TOP**	**8**	**stage B all C**	**0.52**	**32/296**	**nodes 20 mm apart at ring**	**friend-AP8-diagrid-evenload-opentop**
12	AP8 even load 4 rings	8	stage B all C	0.90 (corner legs)	40/224	32 flat bars; stage A tension	friend-AP8-diagrid-evenload-4ring
13	**16 FEET (5/side) + OPEN TOP**	**16**	**A and B all C**	**0.52**	**36/176**	**4 tiny 14 mm bars at ring**	**friend-16-diagrid-evenload-opentop**



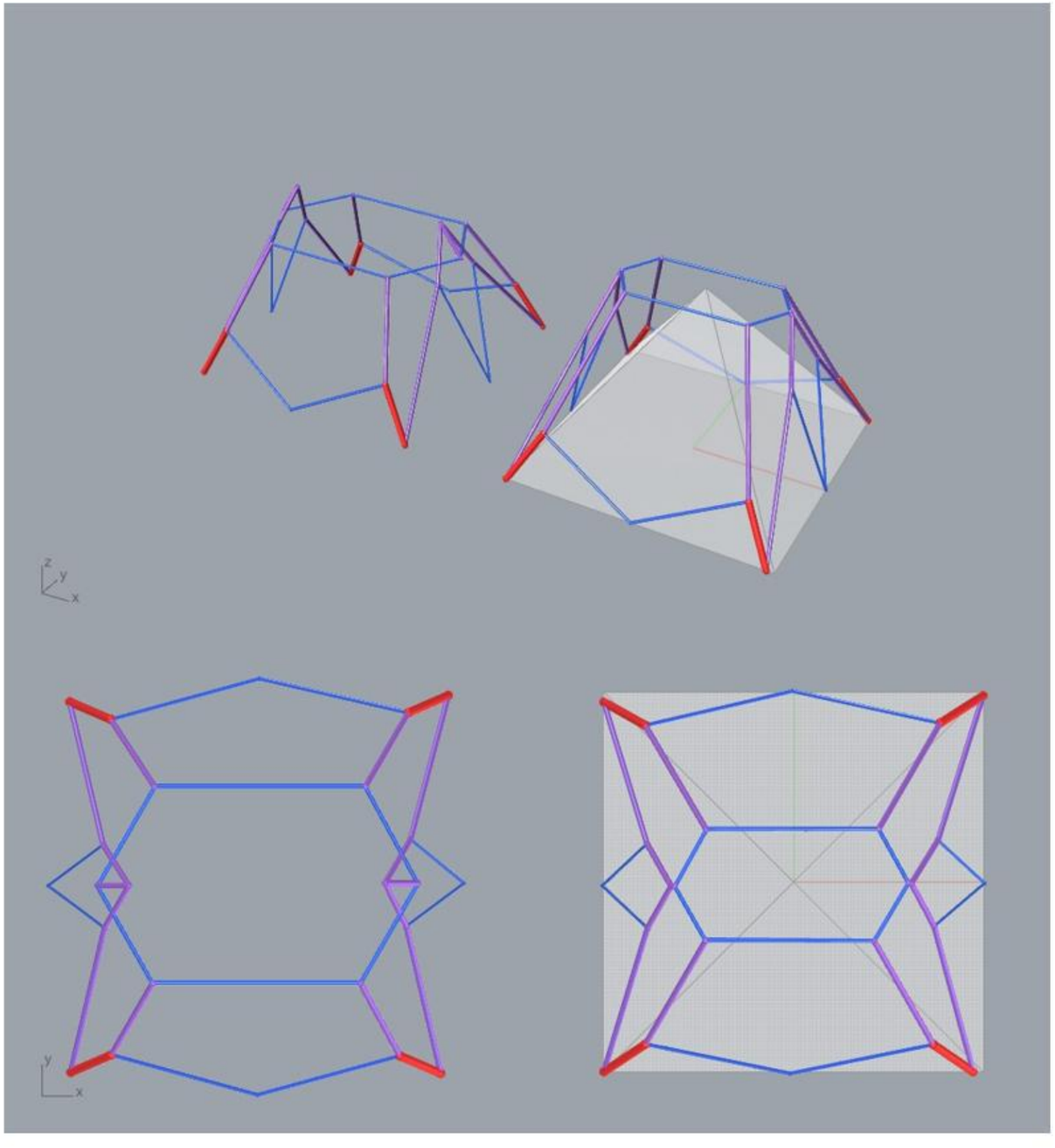
1 Panel A frame



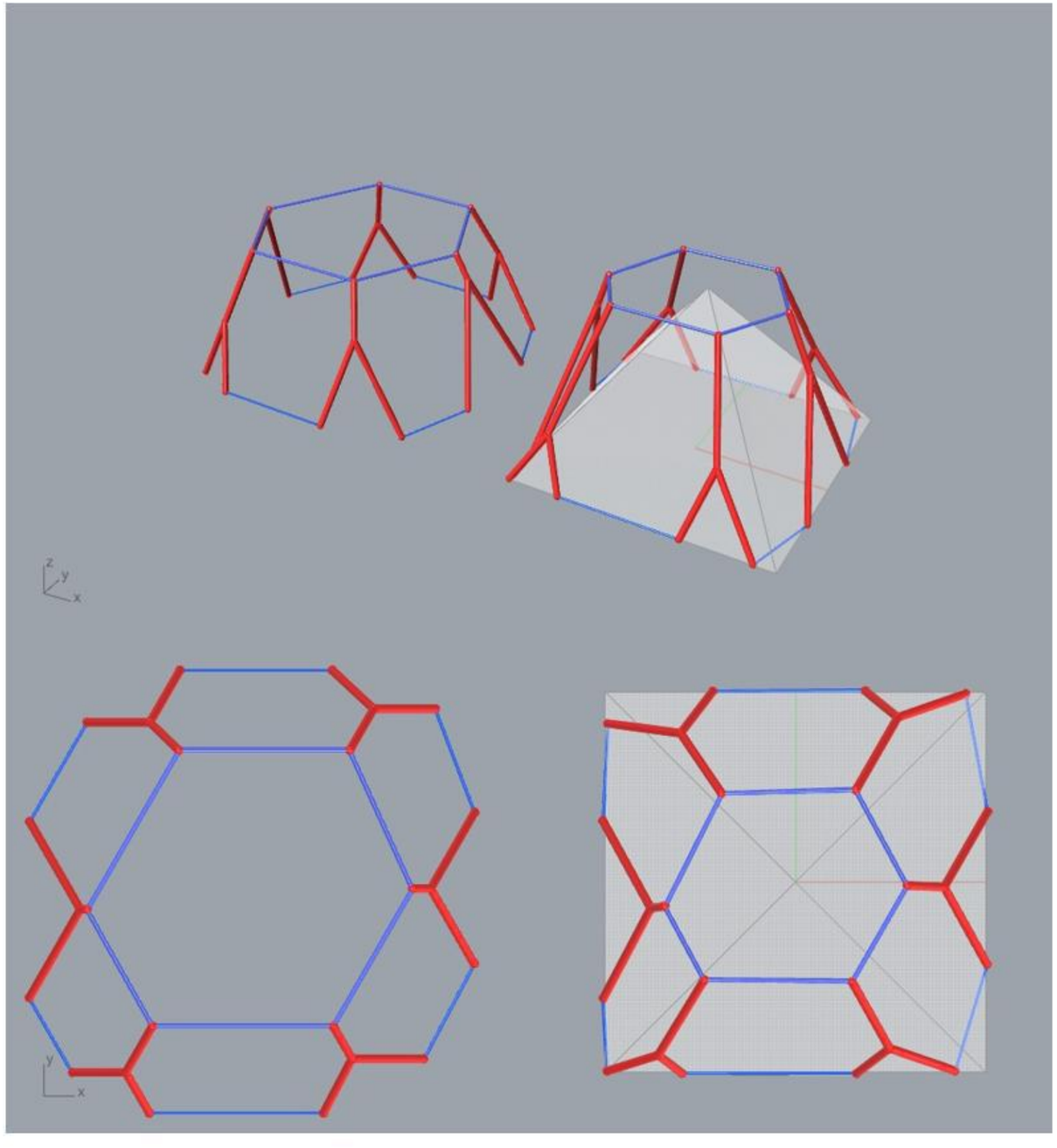
2 Panel A shell (PolyFrame example 3(7))



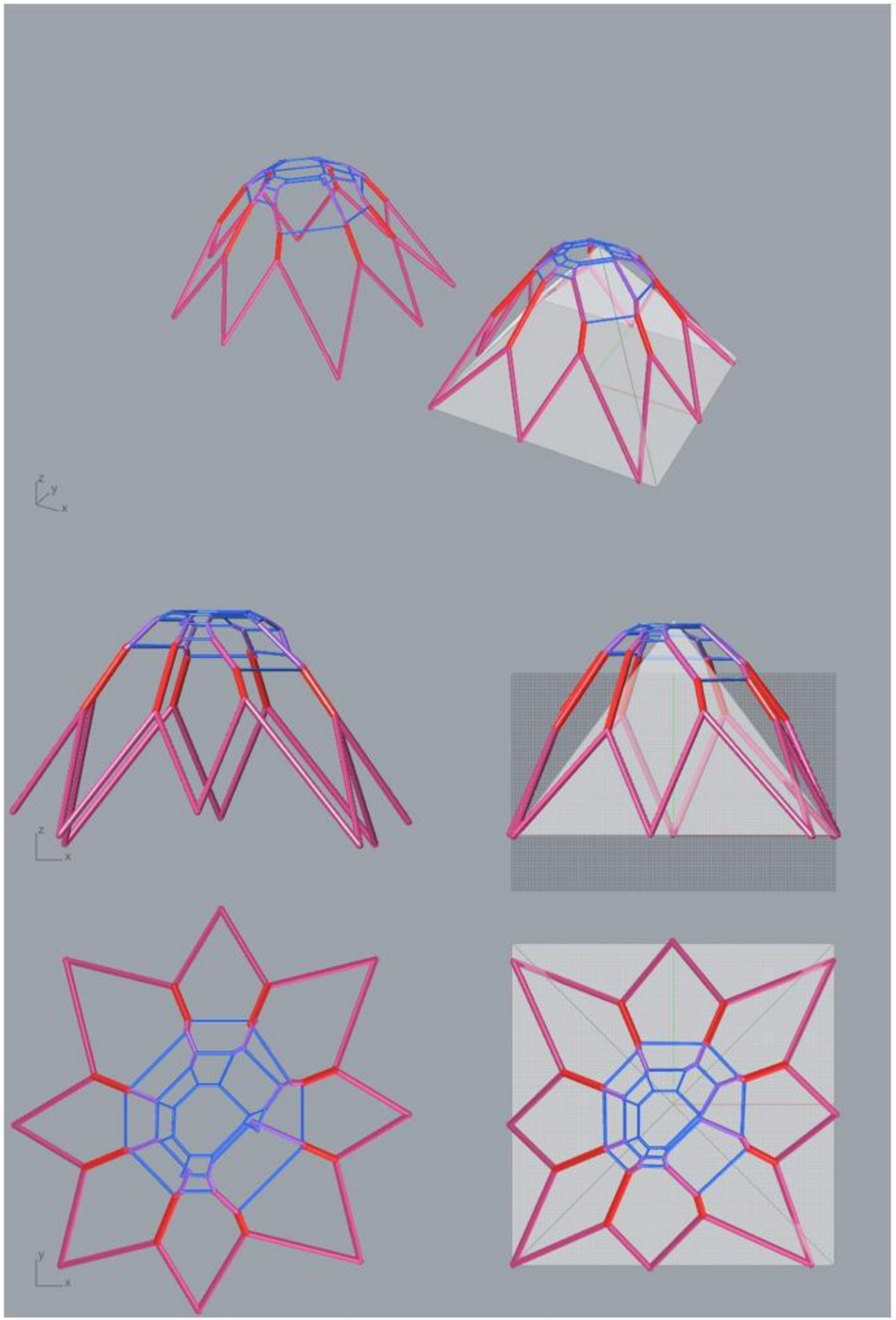
3 AP8 frame



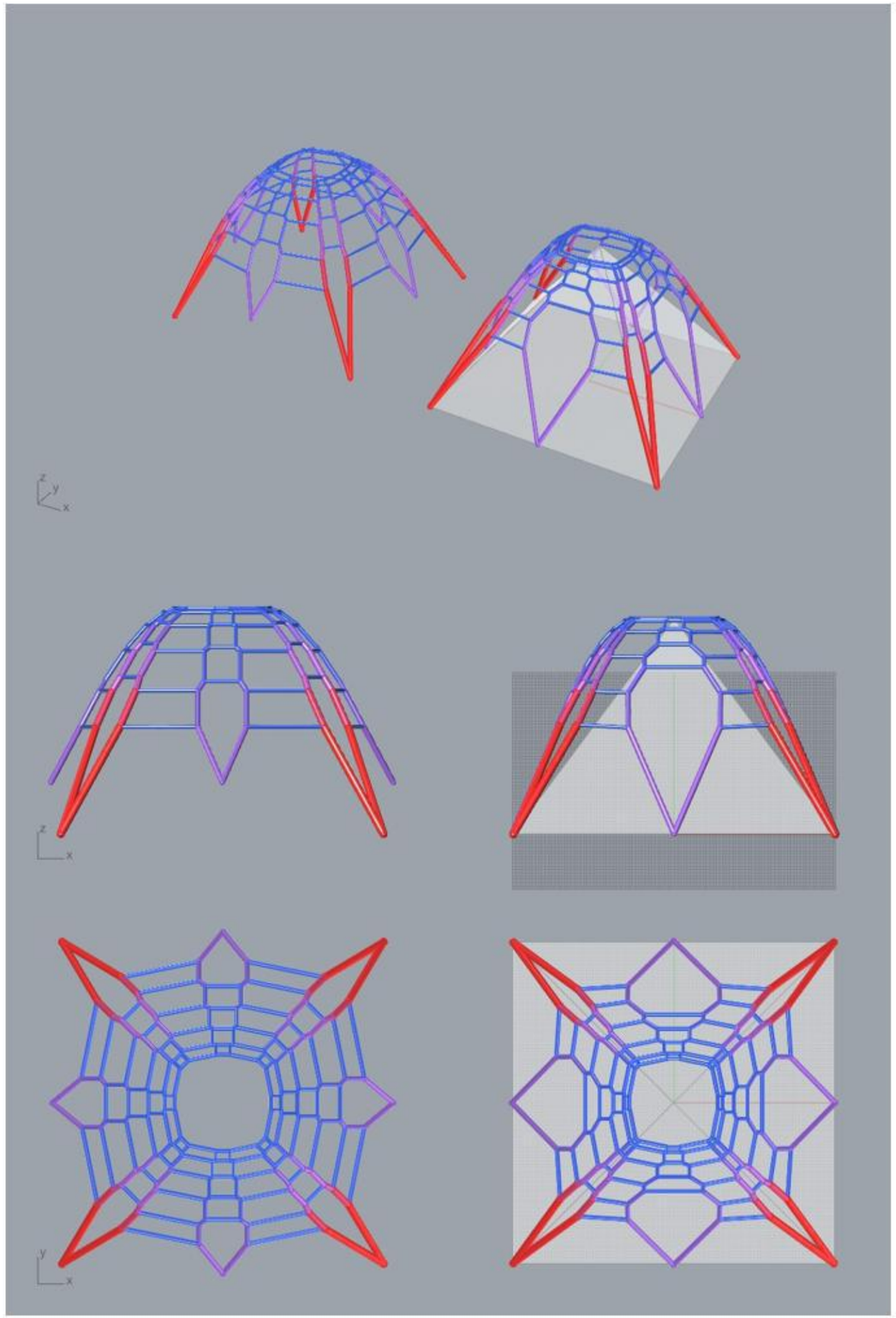
4 AP12 frame



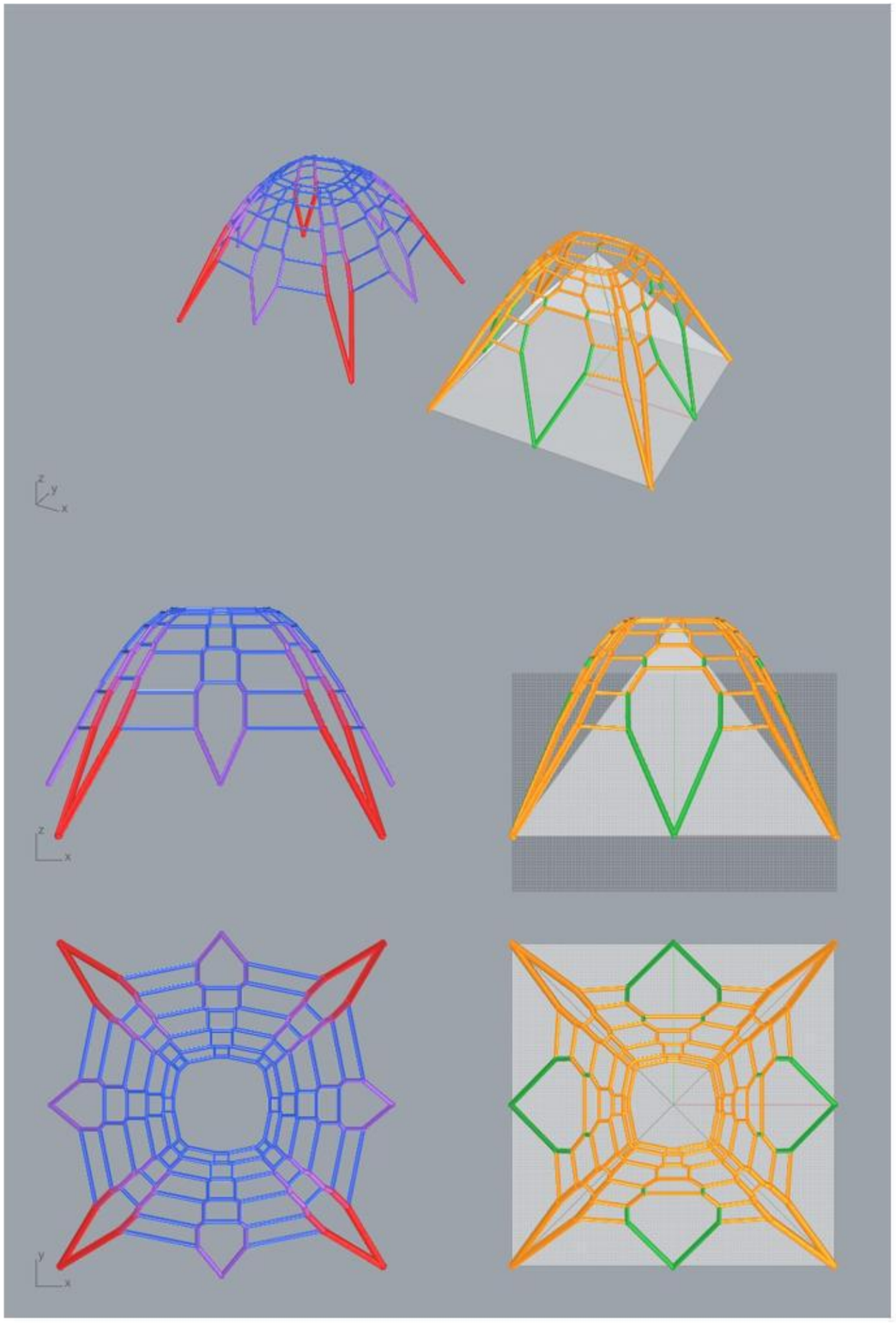
5 AP8 shell GUESS



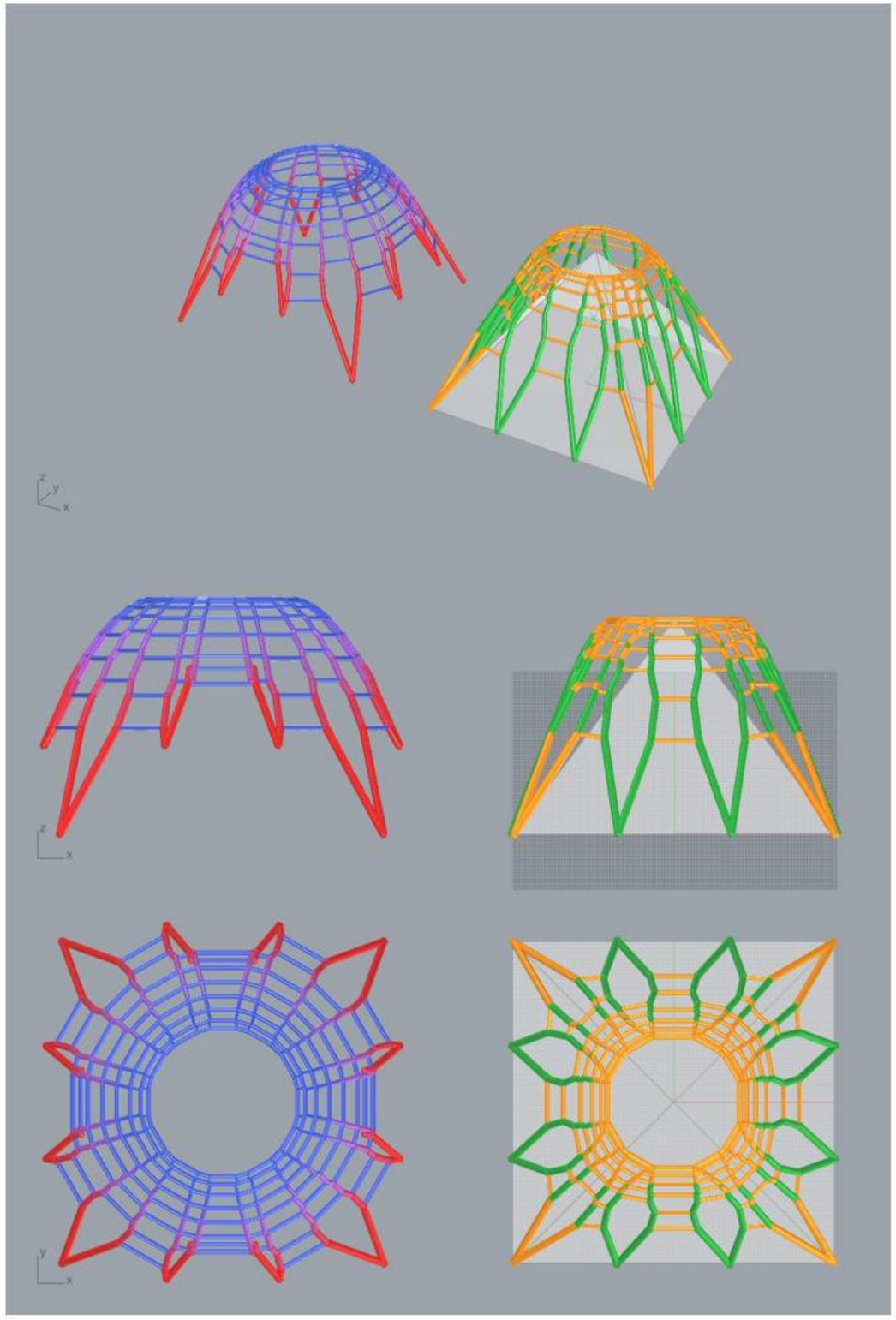
6 AP8 shell v2



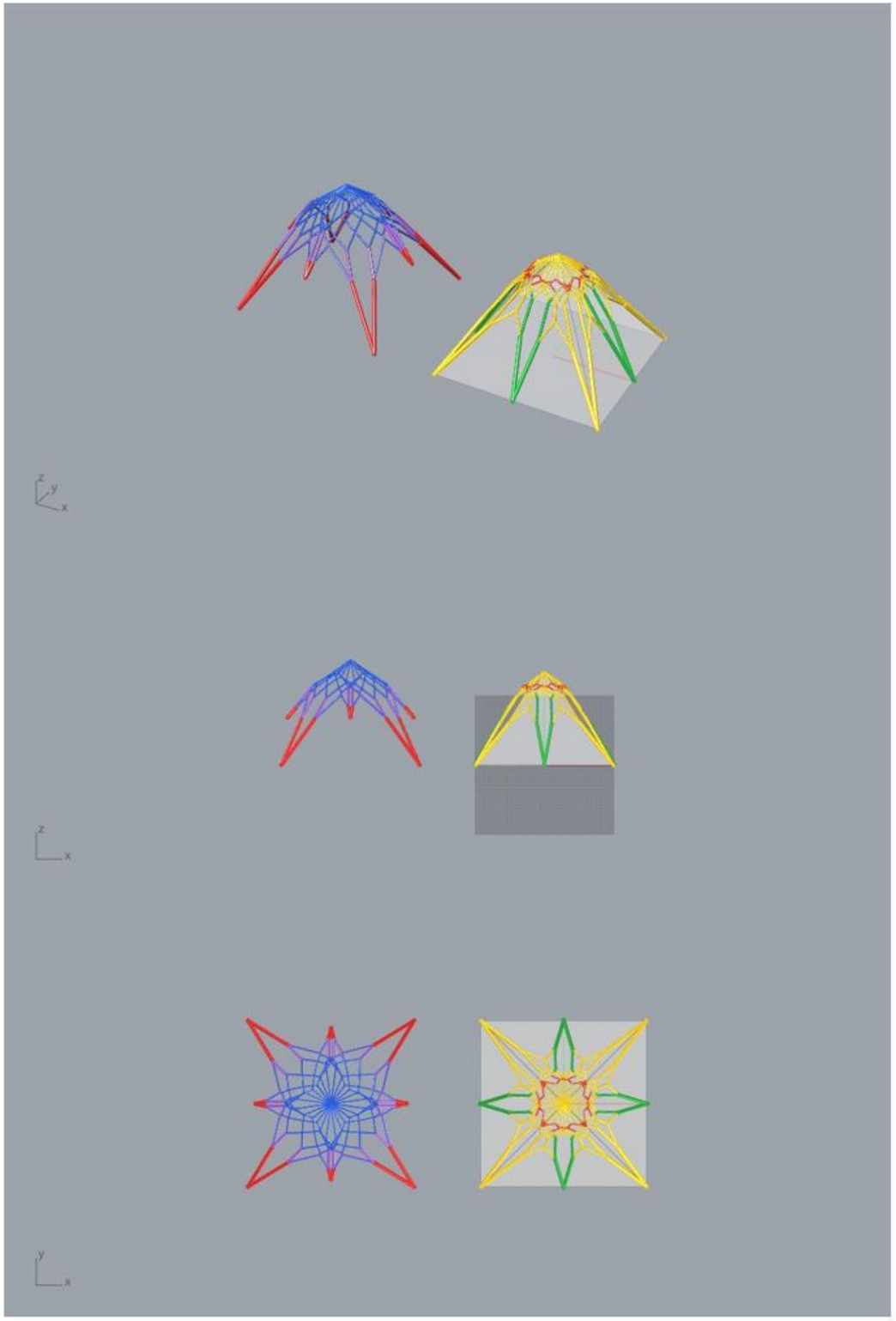
6b AP8 shell v2 - print (green) / cast (orange)



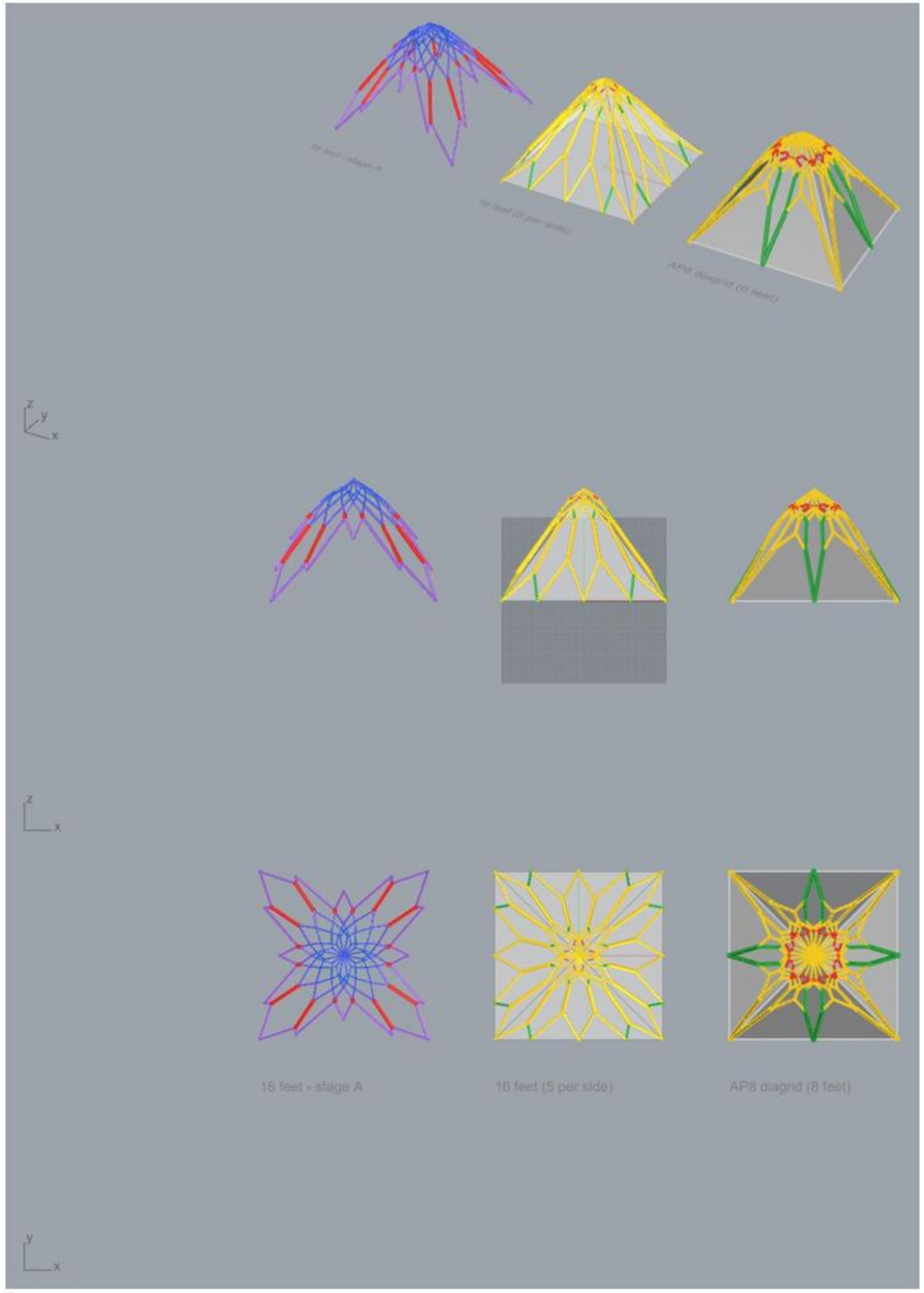
7 AP12 shell



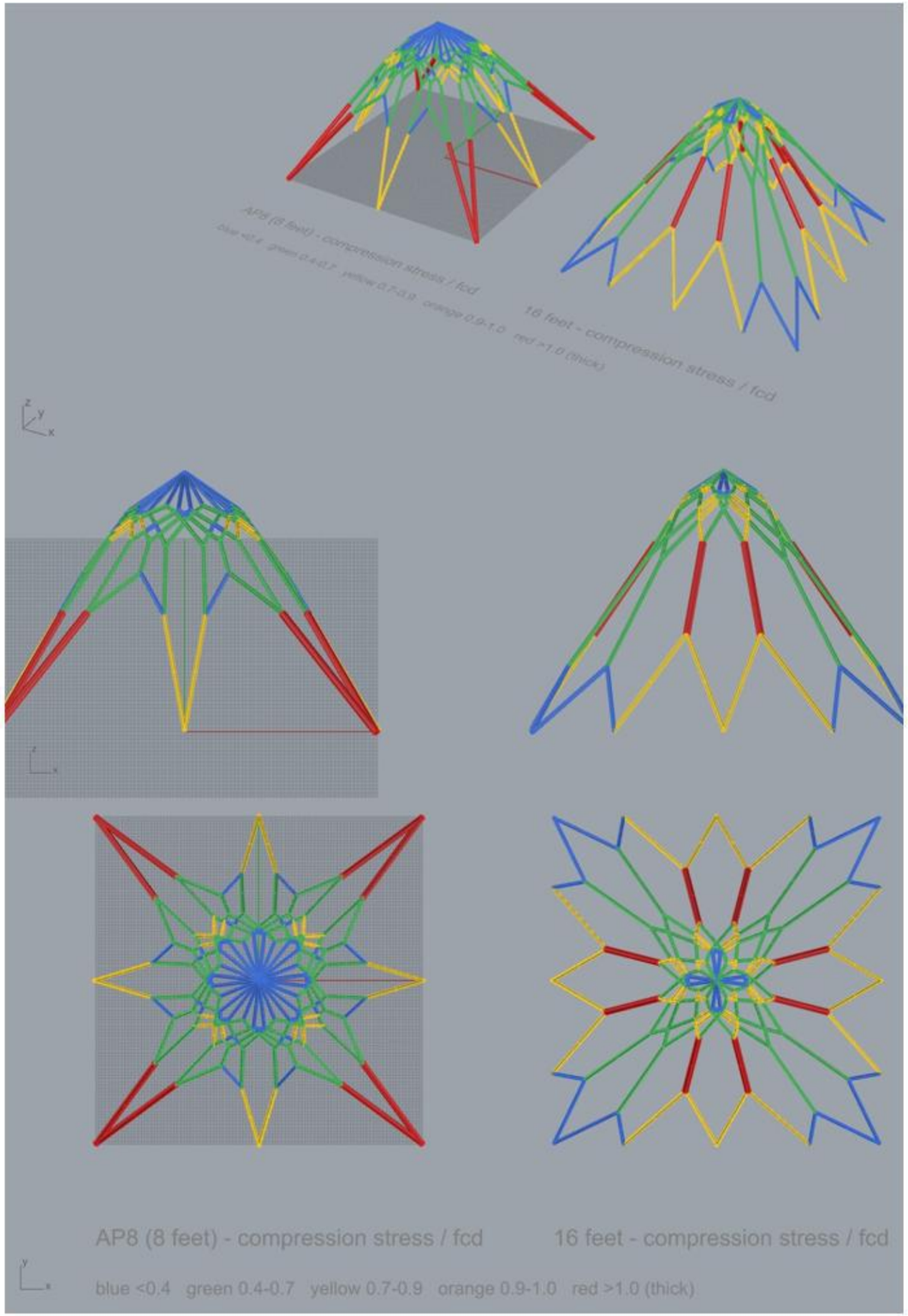
8 AP8 diagrid



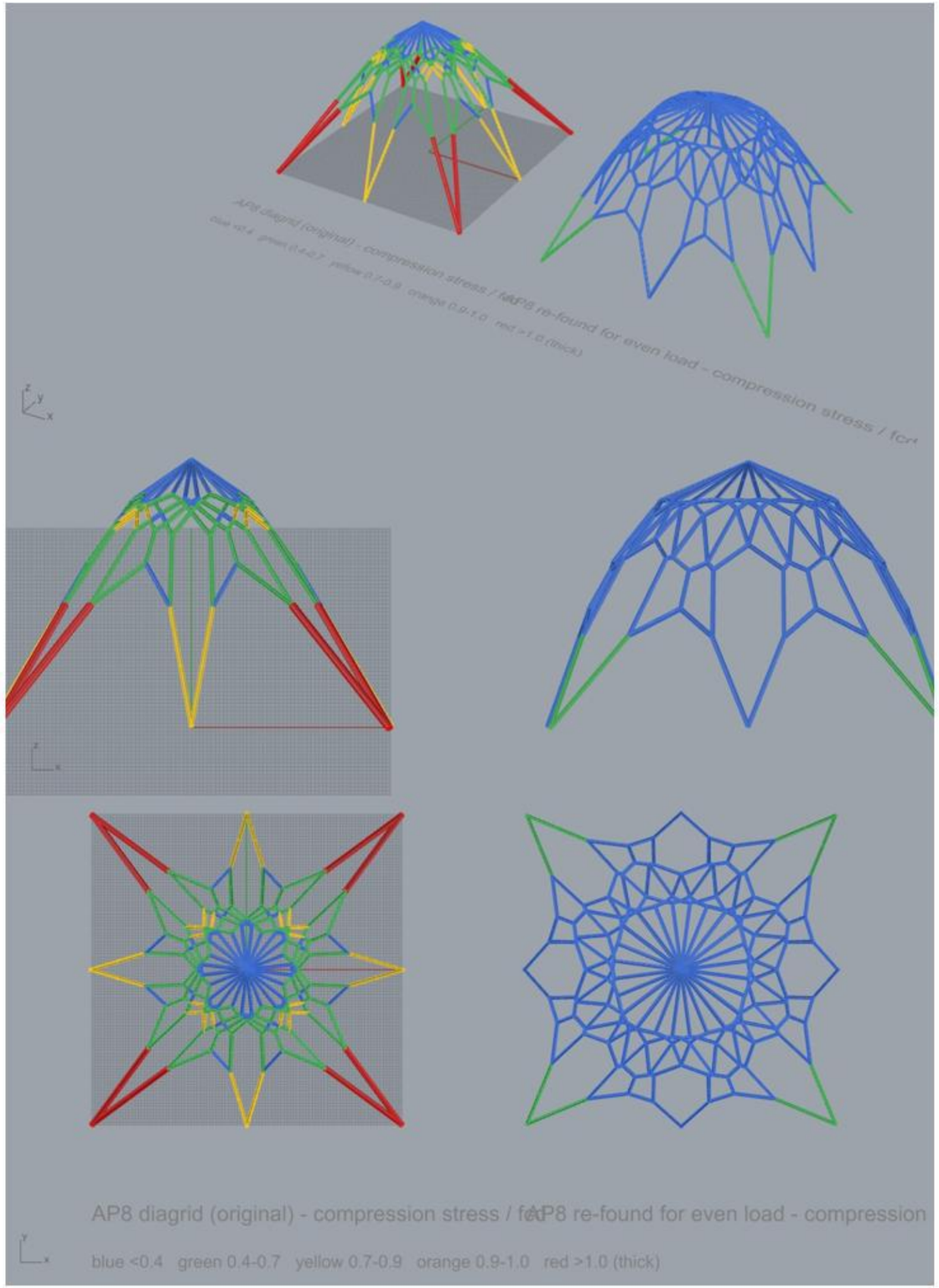
9 16-feet diagrid (5 per side) next to AP8



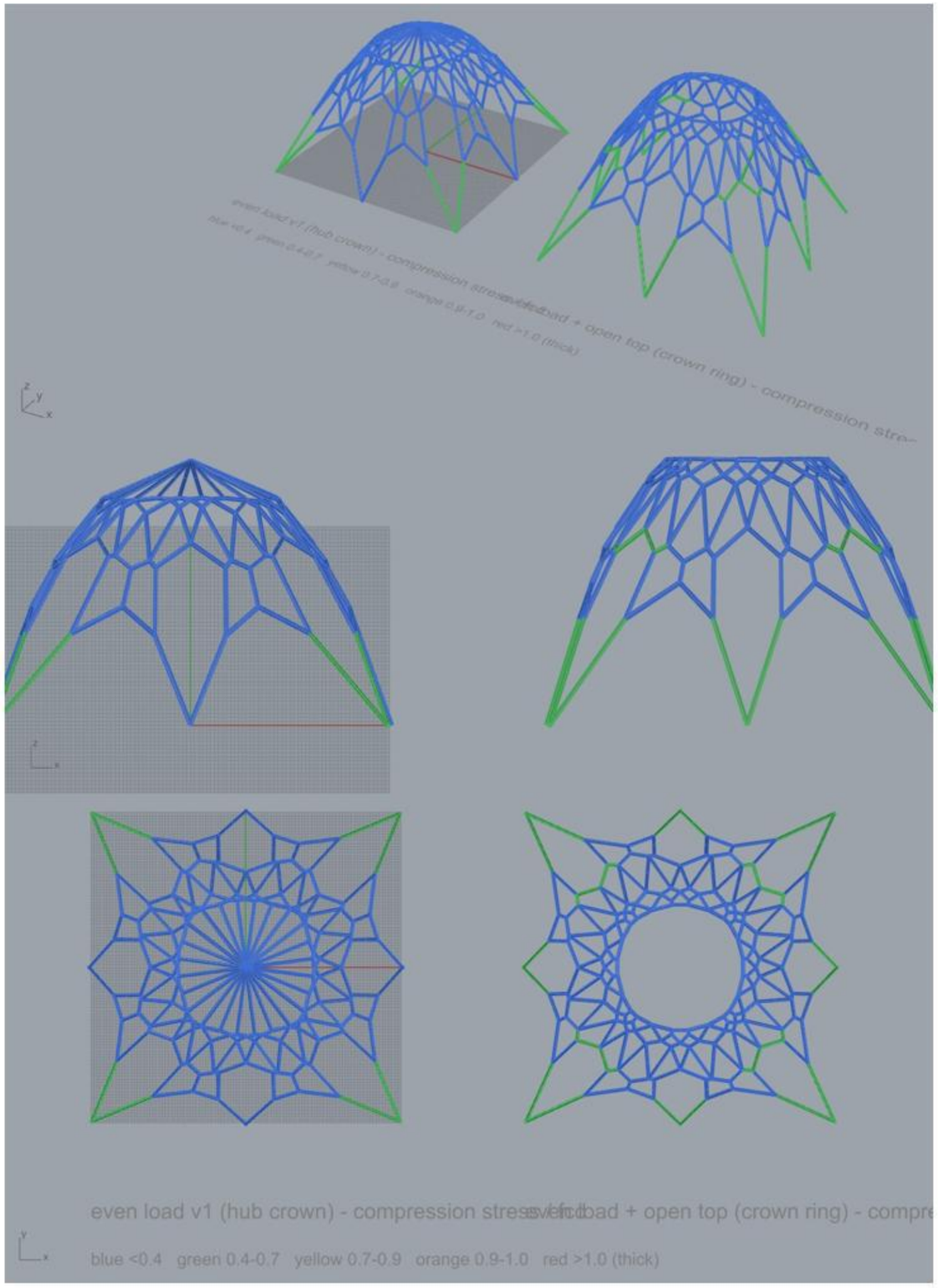
8+9 Karamba check: AP8 diagrid vs 16-feet



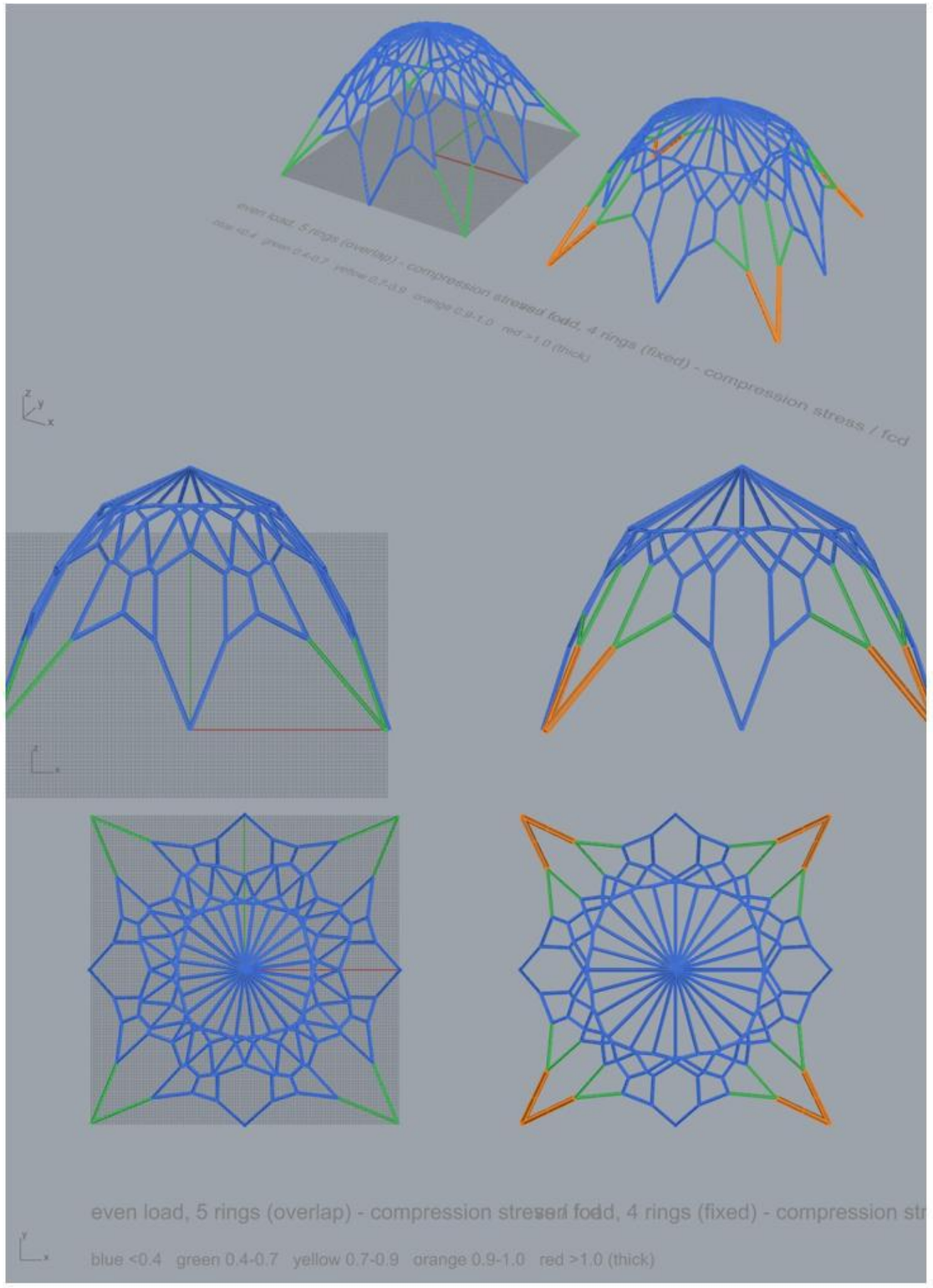
10 AP8 even load (right) vs original (left), Karamba



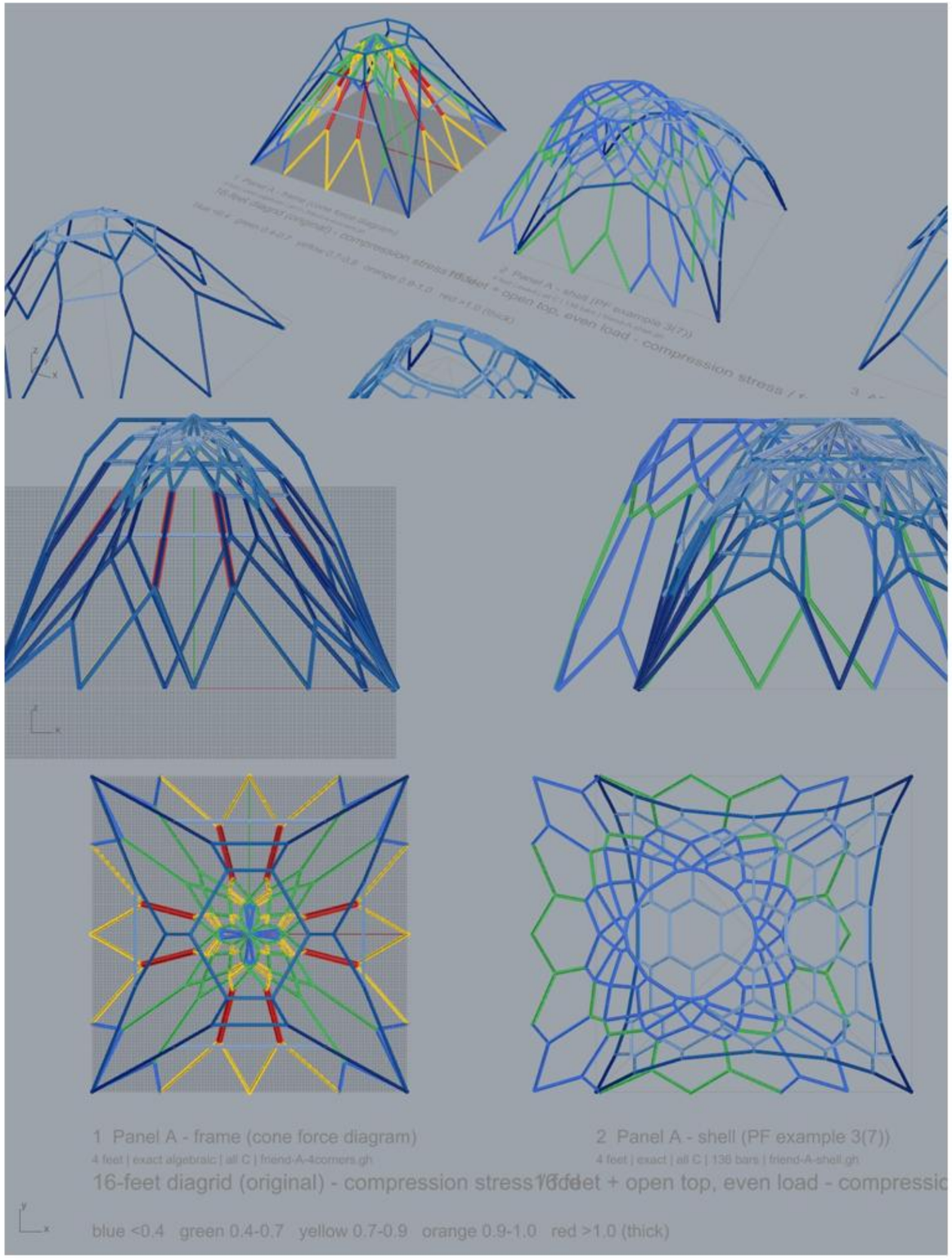
11 OPEN TOP (right) vs even-load hub (left), Karamba



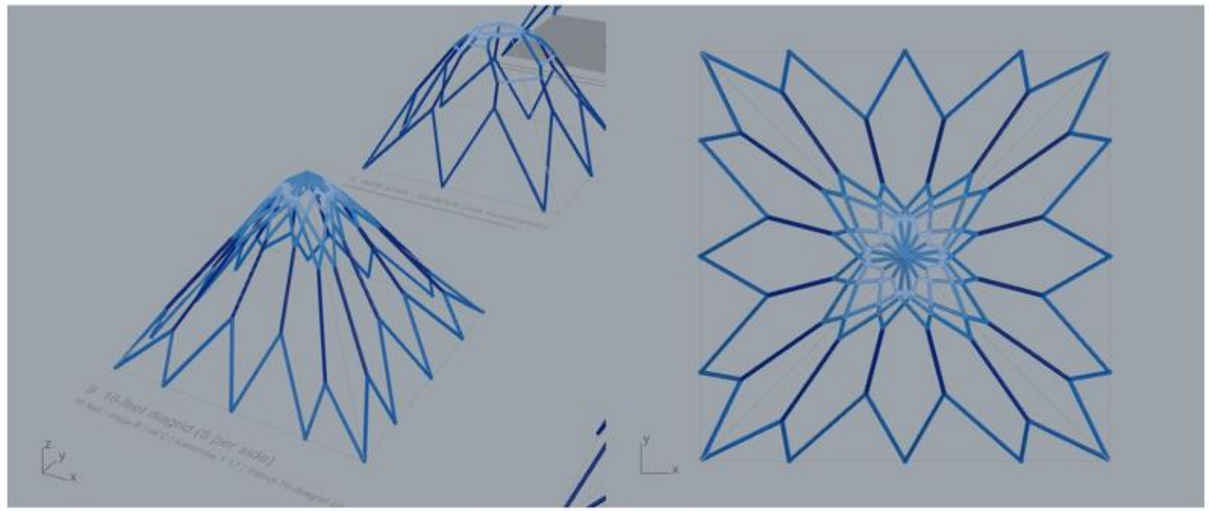
12 4 rings (right) vs 5 rings with overlap (left), Karamba



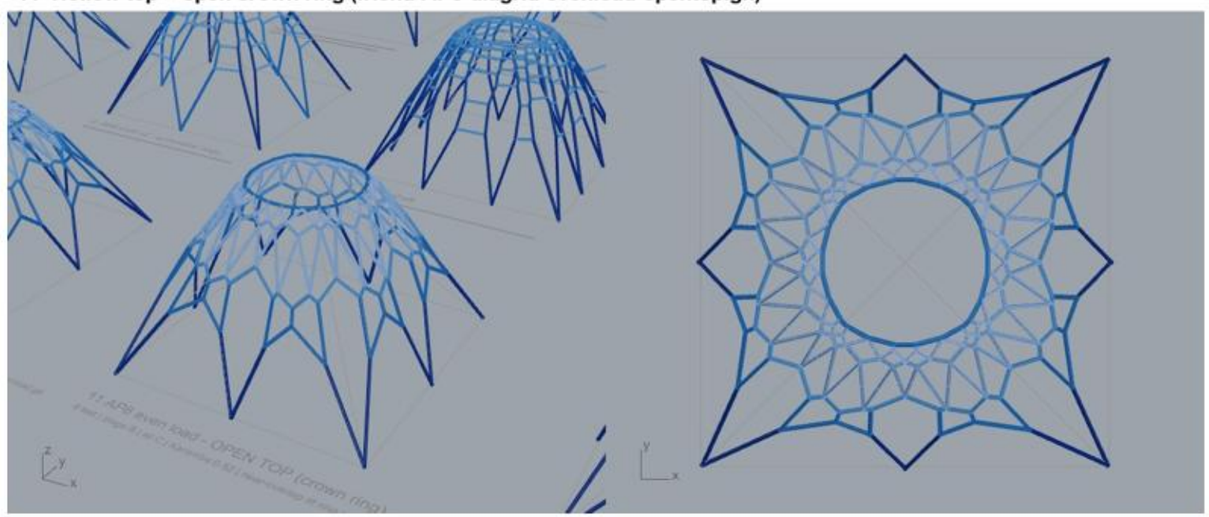
13 16 feet (5 per side) + OPEN TOP, re-found for even load - Karamba vs original 16-feet diagrid (left)



9 16-feet diagrid = 5 supports per side (friend-16-diagrid.gh)



11 Hollow top = open crown ring (friend-AP8-diagrid-evenload-opentop.gh)



Close-ups of #9 (5 per side) and #11 (hollow top) as they appear in the 3dm